ADARSH SHETH

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OBJECTIVE

Aerospace Engineering Junior with hands-on experience in CAD, CFD, 6DoF flight dynamics, hardware testing, mechanical assembly, rapid prototyping, mission analysis, and systems integration in design team and competition settings, seeking a Summer 2027 internship in any location, specializing in structures, mechanical engineering, product design, or traditional aerospace roles.

EDUCATION

Texas A&M University, College Station, TX — Expected May 2028

Bachelor of Science in Aerospace Engineering

- Coursework: Structural Analysis II, Materials Science, Mechanics of Materials, Fluid Dynamics, Control Theory, Vehicle Dynamics, Orbital Mechanics, Thermodynamics, Numerical Simulation, Circuits, Program Design, High Speed Aerodynamics
- Leadership: program coordinator for UpLift (HS drone design program), design team lead for SEDS Lunar Search & Rescue

EXPERIENCE

Aerodynamics Engineer — Sep 2025 - Present

Texas A&M Vertical Flight Design Team — *College Station, TX*

- Executed 100+ parametric CFD analyses (SimScale) for 25-lb hexacopter, feeding forces data into 6DoF dynamics model, ensuring 100% flight stability under 15-kt gusts and autonomous payload pick-up and delivery precision within 3-by-3 ft zone
- Integrated translational and hover state mass-power functions into 1000 flight paths (Monte Carlo) to optimize firefighting response, achieving -27% time and +17% points increase, winning 1st presentation and 2nd technical report at DBVF competition
- Iterated arm-folding mechanism and authored field installation guide for compact transport and 50% faster pre-flight deployment

Systems Engineer Intern — Jun 2026 – Aug 2026

Raytheon — *Woburn, MA*

- Architected satellite acquisition pipeline across 15K+ LEO parse (ECI/ECEF/AER) under FOV/kinematic/RSO signature constraints for Over-the-Horizon Radar (OTHR) modernization initiative, slashing runtime by 50 minutes across benchmarks
- Developed standalone target-tracking script (MATLAB, Bitbucket) with dynamic geospatial trajectory overlay and partner-supplied propagator integration for live customer demonstration, easing radar operations and eliminating external dependencies
- Automated TLE fetch (Space-Track API) and refactored data schemas to Alpha-5 to ensure downstream propagation continuity

Fluid Systems Engineer — Feb 2025 – Jan 2026

Texas A&M Rocket Engine Design Team — *College Station, TX*

- Developed fluid system feedline schematics mapping NPT/JIC/CGA interfaces with control elements for streamlined assembly and formulated 11’ tank precision machining approach complemented by O-ring micro-irregularity absorption for fuel separation
- Executed Upper Nitrous FMEA (RPN weighting) and formulated N2O leak/pressure test procedures (P&ID) for Elysium 2.0 bi-propellant engine, presenting fill process and integration visualizations within V&V framework to clear PDR/CDR milestones
- Automated BoM with cost authentication, part overrides, and dynamic inventory for hardware traceability and financial records

Lead Aircraft Sizing & Performance Engineer — Nov 2025

Aggies Ignite Challenge — *College Station, TX*

- Designed lift+cruise mountain S&R eVTOL, delivering 5-lb payload (10K ft altitude, 40 kt winds, 50% visibility), placing 1st
- Programmed iterative conceptual aircraft sizing algorithm (Python) to evaluate 2M+ configurations against conflicting weight and aerodynamic constraints, optimizing planform area and rotor fractions to cut MTOW by 112 lbs, sustaining cruise at 42 mph
- Validated with sourcing analysis, CAD model (Fusion 360), structural analysis (FEA), and 3D printing slice (Bambu Studio)

Co-Founder, Captain, Mechanical Engineer — Mar 2022 - Jun 2024

Dougherty Valley VEX Robotics — *San Ramon, CA*

- Oversaw CAD (Fusion 360), T&E, machining, mechanical assembly, strategy optimization, documentation, and systems integration for 12-member team, winning 8 awards, 2 world championship qualifiers, and ranking in top 2% of 20,000 teams
- Established chain of command, project management workflow, underclassmen mentorship system, and $9,000 in funding from the ground up, outperforming 4 sister teams and setting a historical achievement of 100% world qualification rate
- Authored 1 technical, 1 management, and 2 operational guides, creating documentation for programs with $30K+ annual revenue

AWARDS, SKILLS, & INTERESTS

- Eagle Scout & 4 Palms, 1st Place Aggies Ignite Challenge, VEX Worlds 2023 & 2024, 8x Robotics Awards, 3X PVSA, CSF
- Autodesk Fusion 360, SolidWorks, Onshape CAD, FEA, DFA, Kinematic Analysis, Assembly, FMEA, GD&T, P&ID, T&E, V&V, SimScale CFD, OpenFOAM, XFOIL, XFLR5, MATLAB, Simulink, Propeller/Rotor Analysis, Flight Dynamics, Vehicle Sizing, Pneumatics, 6DOF, Python, C++, VBA, Git, GitHub/Bitbucket, Monte Carlo Simulation, Numerical Methods, Confluence
- Hiking, Camping, Photography, Editing, Travel, Philosophy/Ethics, Robotics, Catan, Baking, Origami, Digital Design